



Greenhouse Gasses Emissions in World's Economy

Aleksandra Dacko

Data handling and visualization details

All data handling, plotting, and text generation were written in a single R Markdown file and were based on the EDGAR report (Crippa and Pekar 2024). Using R Markdown allowed me to integrate R code, Markdown, and LaTeX to ensure reproducibility and transparency of the analysis process and report generation. I processed raw data by performing filtering, grouping, and deriving new indicators. I used `dplyr` and `tidyr` R libraries to clean, transform, and aggregate the data while visualizations were created using `ggplot2` and `treemapify` R packages. The report format was designed and written in LaTeX explicitly for this exercise. Chart's specific data handling process were as follows:

- **Chart 1:** The year-on-year percentage change in GHG emissions was calculated for the EU27, the Euro area (EA), and World totals. Emissions for the Euro area were derived by summing the totals of the countries within the EA. The results were presented on a line chart with visible data points.
- **Chart 2:** Data on income categories and total populations was sourced from the publicly available World Bank database. After cleaning the data (re-coding country codes and removing incomplete entries) to synchronize it with the EDGAR data, the emissions per capita were calculated for each income group for the year 2023. The results were visualized in a simple bar chart.
- **Chart 3:** Country to continent mapping data was collected. The percentages of total global GHG emissions were calculated for each of the six continents, as well as for individual countries. The results were displayed using a nested treemap plot.

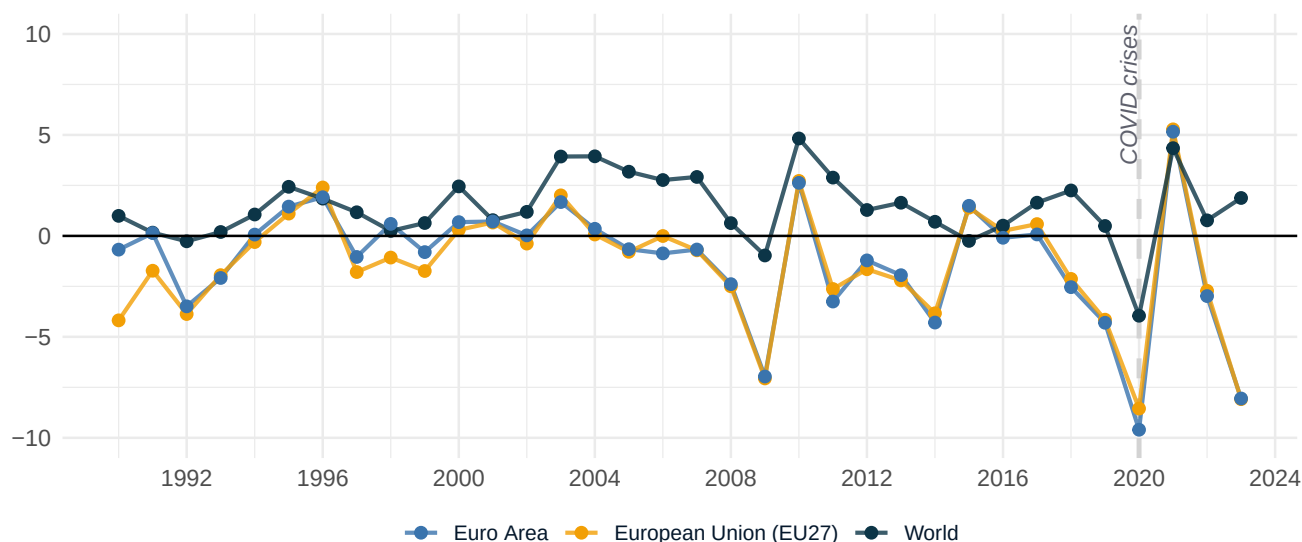
The report's source code can be found in my GitHub repository (<https://github.com/a-dacko/EDGAR-GHG-Emissions-report>).

Evolution of GHG growth in the Euro Area, European Union (EU27) and worldwide

- In the European Union (EU27) and the Euro Area, there was a noticeable **8.1%** decrease in total GHG emissions from 2022 to 2023, amounting to **261 Mt CO_{2-eq}** and **196 Mt CO_{2-eq}** respectively. On the other hand, the World GHG emission increased by **1.8%** (**994 Mt CO_{2-eq}**) between the same time frame.
- From 1990 to 2023 the EU27 and Euro Area countries showed steady decline in GHG emissions with compounded annual growth rate (CAGR) of **-1.2%**. In contrast, total World GHG emissions continue to raise with CAGR of **1.5%**.

Greenhouse Gasses Emission – Year on year growth rate

Growth rate in %



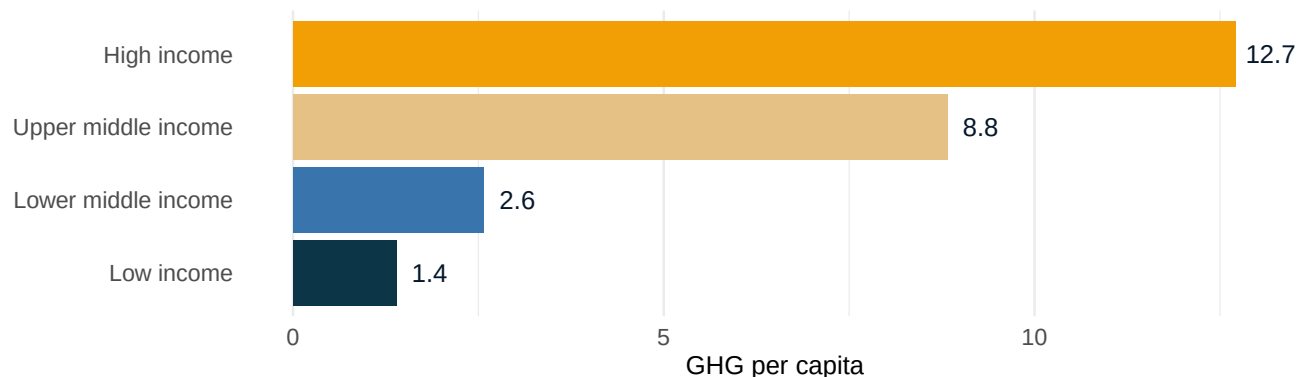


Comparison of countries' GHG emissions per capita aggregated according to the World Bank income groups

- There is disproportionate emissions responsibility of high-income and upper-middle-income nations, leading to high emissions per capita scores for those groups.
- High-income countries emit around **12.4 t CO₂-eq** per capita, whereas low-income countries emit almost eight times less, at **1.5 t CO₂-eq** per capita.
- There is a strong relationship between a country's Gross Domestic Product and the amount of GHG it produces (Hamed Bin Furkan and Uddin 2023). Wealthier countries, due to their advanced industrialization and economic growth, produce significantly more GHG emissions.

Greenhouse Gasses Emission per capita – 2023 income groups comparison

Tonnes of CO₂-eq per capita per year



Notes.

(1) Individual countries were classified based on the latest classification provided by World Bank

(<https://datahelpdesk.worldbank.org/knowledgebase/articles/378834-how-does-the-world-bank-classify-countries>).

(2) The GHG per capita was calculate using population totals from 2023 provided by World Bank (<https://data.worldbank.org/indicator/SP.POP.T>).

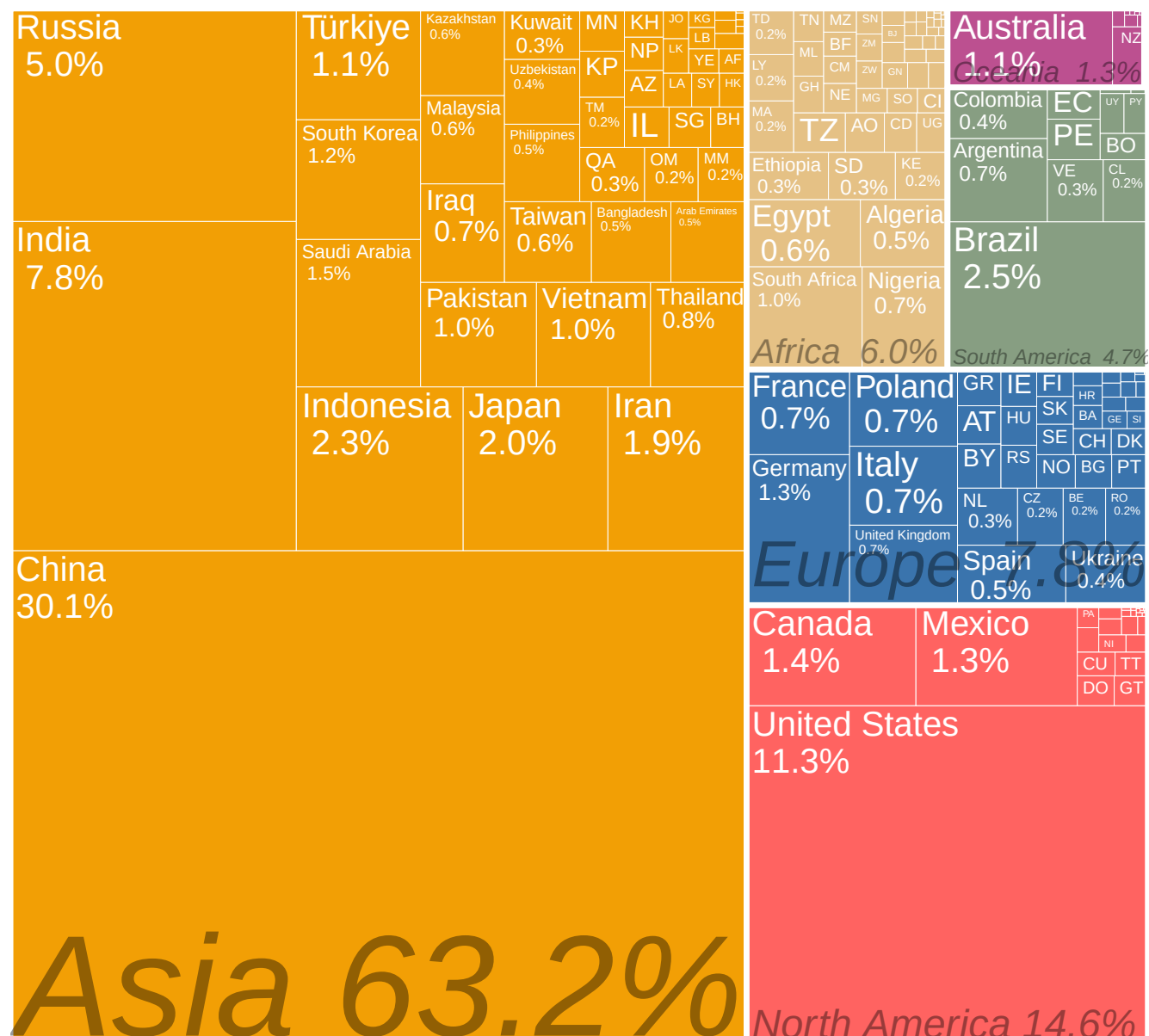


Contribution of individual country and continent GHG emissions to total World GHG emissions

- China, the United States, India, Russia, and Brazil were the world's largest GHG emitters in 2023. Together, they accounted for 49.8% of the global population and **62.7%** of global GHG emissions.
- Asia is responsible for the largest share of world emissions due to rapid industrialization in countries like China and India. On the contrary, Africa, despite having the 2nd largest share of the world's population (after Asia), accounts for only **6.0%** of total global GHG emissions.

Contribution of continent and individual country GHG emissions to total World GHG emissions in 2023

Proportion of World total GHG emissions in %



Crippa, Guizzardi, M., and F. Pekar. 2024. "GHG Emissions of All World Countries." *Publications Office of the European Union, Luxembourg*. <https://doi.org/10.2760/4002897>.

Hamed Bin Furkan, Rakibul Hasan, Kazi Md, and Md Jamal Uddin. 2023. "Greenhouse Gas Emission, GDP, Tertiary Education, and Rule of Law: A Comparative Study Between High-Income and Lower-Middle Income Countries." *Heliyon*. <https://doi.org/https://doi.org/10.1016/j.heliyon.2023.e16265>.